3 Background

In this section, we introduce our problem setup within the framework of a decentralized partially observable Markov decision process (Dec-POMDP) (Oliehoek & Amato, 2016; Amato, 2024). We describe how agents operate under partial observability and must make decisions based on local observations and histories. We then discuss the episodic multi-agent reinforcement learning (MARL) setting, where agents receive rewards only at the end of each episode, posing a significant challenge for credit assignment. To address this, we explore potential-based reward shaping for multi-agent systems (Ng, 1999; Devlin & Kudenko, 2011), a technique that reshapes the reward function to facilitate learning while preserving the optimal policy. Finally, we analyze how imperfect credit assignment impacts the variance of the policy gradient in multi-agent systems. We show that improper credit distribution among agents leads to high variance in advantage estimates, which in turn exacerbates the learning process and hinders the convergence to optimal policies.

3.1 Decentralized Partially Observable Markov Decision Processes (Dec-POMDP)

A Dec-POMDP is represented by a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{T}, \mathcal{O}, \mathcal{N}, \mathcal{R}_\zeta, \rho_0, \gamma)$ where $s \in \mathcal{S}$ is the environment state space, $a \in \mathcal{A}$ is the joint action space denoted by $\mathcal{A} := \mathcal{A}_1 \times \mathcal{A}_2 \times ... \times \mathcal{A}_N$ and $\mathcal{P} : \mathcal{S} \times \mathcal{A} \times \mathcal{S} \to [0, 1]$ is the state transition function. $r_{\text{global},t} \sim \mathcal{R}_\zeta(|s_t, a_t) : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the global reward shared among agents at every timestep of the trajectory. ρ_0 is the initial state distribution and $\gamma \in [0, 1]$ is the discount factor. $\pi = \prod_{i=1}^{N} \pi_i$ is the joint policy of the multi-agent system which comprises of independent agent policies π_i. Each agent $i \in \{1 \dots N\}$ receives an observation $o_i \in \mathcal{O}_i$ from the observation function $\mathcal{T}(s, i) : \mathcal{S} \times \mathcal{N} \to O$. Because the state is not directly observable, it is typically beneficial for each agent to remember a history of its observations or observations-actions. $h_t \in \mathcal{H} := \mathcal{H}_1 \dots \mathcal{H}_N$ is the set of agent observation (-action) histories up to the current time step t where $h_{i,t} \in \mathcal{H}_i$ and defined as $h_{i,t} = \{o_{i,1}, a_{i,1}, ..., o_{i,t}\}$ denotes agent i's history and $h_{\neg i,t}$ is the history of all other agents except agent i. At each time step every agent selects an action $a_i \in \mathcal{A}_i$ according to it's policy $\pi_i : H_i \times \mathcal{A}_i \to [0, 1]$. $\tau = \{o_{0,1}, a_{0,1}, ...o_{0,N}, a_{0,N} .....o_{|\tau|,1}, a_{|\tau|,1} ...o_{|\tau|,N}, a_{|\tau|,N}\}$ is the multi-agent trajectory where $|\tau|$ is the horizon length of the trajectory. The goal of the agents is to determine their individual optimal policies that achieve maximum global return $E_{s_0 \sim \rho_0, s \sim \mathcal{P}, a_i \sim \pi_i ..} \left[\sum_{t=1}^{|\tau|} \gamma^t r_t \right]$.

3.2 Return Decomposition in Episodic Multi-Agent Reinforcement Learning

In most MARL systems, each agent receives a global reward $r_{\text{global},t}$ after executing the joint action a_t in state s_t. However, in episodic MARL setups, agents only receive a global reward signal from the environment at the end of the trajectory, known as the episodic reward or trajectory return $r_{\text{global,episodic}}$. The objective in such environments is to maximize the trajectory return, $E_{s_0 \sim \rho_0, s \sim \mathcal{P}, a_i \sim \pi_i ..}(r_{\text{global,episodic}}(\tau))$. Delayed reward settings introduce significant bias and variance (Ng, 1999) during the learning process, exacerbating sample inefficiency.

3.3 Potential-based reward shaping

Ng (1999) presented a single-agent reward shaping method to address the credit assignment problem by introducing a potential-based shaping reward to the environment. The combination of the shaping reward with the original reward can enhance the learning performance of a reinforcement learning algorithm and accelerate the convergence to the optimal policy. Devlin & Kudenko (2011) and Xiaosong Lu (2011) extended potential-based reward shaping to multi-agent systems.

Theorem 1. *Given an n-player discounted stochastic game $M = (S, A_1, \dots, A_n, T, \gamma, R_1, \dots, R_n)$, we define a transformed n-player discounted stochastic game $M' = (S, A_1, \dots, A_n, T, \gamma, R_1 + F_1, \dots, R_n + F_n)$, where $F_i \in S \times S$ is a shaping reward function for player i. We call F_i a potential-based shaping function if F_i has the form:*

$$F_i(s, s') = \gamma\Phi_i(s') - \Phi_i(s),$$

where $\Phi_i : S \to \mathbb{R}$ is a potential function. Then, the potential-based shaping function F_i is a necessary and sufficient condition to guarantee the Nash equilibrium policy invariance such that:

- ***(Sufficiency)*** *If F_i $(i = 1, \ldots, n)$ is a potential-based shaping function, then every Nash equilibrium policy in M' will also be a Nash equilibrium policy in M (and vice versa).*
- ***(Necessity)*** *If F_i $(i = 1, \ldots, n)$ is not a potential-based shaping function, then there may exist a transition function T and reward function R such that the Nash equilibrium policy in M' will not be the Nash equilibrium policy in M.*

In summary, potential-based reward shaping ensures that Nash equilibrium policies are preserved, enhancing learning without altering the strategic dynamics. This principle underpins our proposed reward redistribution method, which we will validate in the following sections, demonstrating its effectiveness in multi-agent reinforcement learning.

3.4 Impact of Faulty Credit Assignment on Policy Gradient Variance

To understand the impact of imperfect credit assignment, we analyze the effect of other agents on the policy gradient update of agent i. Consider an actor-critic gradient estimate for a multi-agent system in a Dec-POMDP setting, computed using a state-action sample from an arbitrary timestep t. We make no assumptions about the policy parameters of the agents in the multi-agent system. Ideally, the policy gradient update for agent i should be computed using

$$\hat{\nabla}_{\theta_i} J(\theta, h) = \nabla_{\theta_i} \log \pi_i(a_i|h_i) \mathbb{E}_{\neg h_i, \neg a_i} \left[A(h, a)\right]$$

Computing $\mathbb{E}_{\neg h_i, \neg a_i} \left[A(h, a)\right]$ ischallenging to compute due to the high dimensionality, dependency on other agents, and the computational complexity involved in accurately modeling and estimating the interdependent histories and actions of multiple agents. However, in practice multi-agent policy-gradient algorithms like MAPPO (Yu et al., 2022), MADDPG (Lowe et al., 2017) etc employ $A(h, a)$ to compute the policy gradient update for each agent. As a result, the credit assignment problem manifests as high variance in advantage estimates, leading to slower learning because of noisier policy gradient estimates.

Multi-agent policy gradient methods approximate the true *advantage* by computing $\hat{A}$, which is actually a stochastic advantage estimation of taking a joint action a while observing the joint agent history h, and following the joint policy π. The advantage function is typically defined as $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$, where $Q^\pi(s, a)$ and $V^\pi(s)$ are the state-action value function and state-value function, respectively (Sutton & Barto, 1998). However, in practice the state-action value function and state-value function are approximated using $\hat{Q}^\pi(h, a)$ and $\hat{V}^\pi(h)$ in Dec-POMPDs and there are many ways to compute $\hat{A}$, generally all involving some error, as the true value functions are unknown (Sutton & Barto, 1998; Schulman et al., 2015).. Intuitively, this is the centralized advantage function which measures how much better it is to select a joint action a than a random action from the joint policy π, while in state s. Besides, to update the policy of agent i, we need to compute the advantage of selecting action a_i, taking into account the specific contribution and context of agent i within the multi-agent system. Perfect credit assignment would be possible if the advantage function could be computed perfectly for each agent, as it directly measures how a particular action of an agent impacted the total reward obtained by the group.

The conditional variance of $\mathrm{Var}(\hat{\nabla}_\theta J|h, a)$, given h and a, is proportional to the variance of $\hat{A}$. While this statement typically implies multiple samples are considered to estimate the variance accurately, we can gain insight into the variance introduced by the contributions of other agents by initially

focusing on a single sample scenario.

$$\mathrm{Var}(\hat{\nabla}_\theta J|h,a) = (\nabla_\theta \log \pi(a_i|h_i))\,(\nabla_\theta \log \pi(a_i|h_i))^T\,\mathrm{Var}(\hat{A}|h,a).$$

It is therefore evident that the variance of the policy gradient update is directly proportional to the variance of the advantage estimate: $\mathrm{Var}(\hat{\nabla}_\theta J|h,a) \propto \mathrm{Var}(\hat{A}|h,a)$.
Let us analyze the variance of the advantage function in cooperative multi-agent setting,

$$\mathcal{A}(h,a) = \mathcal{Q}(h,a) - \mathcal{V}(h)$$

Let us assume that agent i's reward contribution at an arbitrary time-step t is denoted by $r_{i,t}$ and the episodic reward described in subsection 3.2 can be derived using $r_{\text{global,episodic}}(\tau) = \sum_{t=1}^{|\tau|}\sum_{i=1}^{N} r_{i,t}|h,a$. From this we can rewrite $\mathcal{Q}(h,a)$ and $\mathcal{V}(h)$ as $\mathcal{Q}(h,a) = \mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{i=1}^{N} r_{i,t}|h,a]$ and $\mathcal{V}(h) = \mathrm{E}_\pi[\mathcal{Q}(h,a)]$

$$\mathcal{A}(h,a) = \mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{i=1}^{N} r_{i,t}|h,a] - \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{i=1}^{N} r_{i,t}|h]]$$

Based on the linearity of expectations on $\sum_{t=1}^{|\tau|}\sum_{i=1}^{N} r_{j,t} = \sum_{t=1}^{|\tau|} r_{i,t} + \sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}$

$$\begin{aligned}\mathcal{A}(h,a) = {} & (\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h,a] + \mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h,a]) \\ & - (\mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h]] + \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h]])\end{aligned}$$

$$\begin{aligned}\mathcal{A}(h,a) = {} & (\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h,a] - \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h]]) \\ & - (\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h,a] - \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h]])\end{aligned}$$

The advantage estimate considering only the contribution of agent i is $\mathcal{A}_i = \mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h,a] - \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|} r_{i,t}|h]]$ the only advantage term that should be considered while calculating the policy gradient update for agent i whereas the advantage estimate due to other agent contributions $\mathcal{A}_{\neg i} = \mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h,a] - \mathrm{E}_\pi[\mathrm{E}_{s_0\sim\rho_0,s\sim\mathcal{P},a_i\sim\pi_i..}[\sum_{t=1}^{|\tau|}\sum_{j\neq i}^{N} r_{j,t}|h]]$ act as noise. Thus,

$$\mathcal{A}(h,a) = \mathcal{A}_i(h,a) + \mathcal{A}_{\neg i}(h,a)$$

Using variance of the sum of random variables,

$$\mathrm{Var}(\mathcal{A}(h,a)) = \mathrm{Var}(\mathcal{A}_i(h,a)) + \mathrm{Var}(\mathcal{A}_{\neg i}(h,a)) + 2\mathrm{Cov}(\mathcal{A}_i(h,a), \mathcal{A}_{\neg i}(h,a))$$

To express the equation in terms of variance, we use the Cauchy-Schwarz inequality, which states that for any two random variables $\mathcal{A}_i$ and $\mathcal{A}_{\neg i}$:

$$\mathrm{Cov}(\mathcal{A}_i(h,a), \mathcal{A}_{\neg i}(h,a)) \leq \sqrt{\mathrm{Var}(\mathcal{A}_i(h,a))\mathrm{Var}(\mathcal{A}_{\neg i}(h,a))}$$